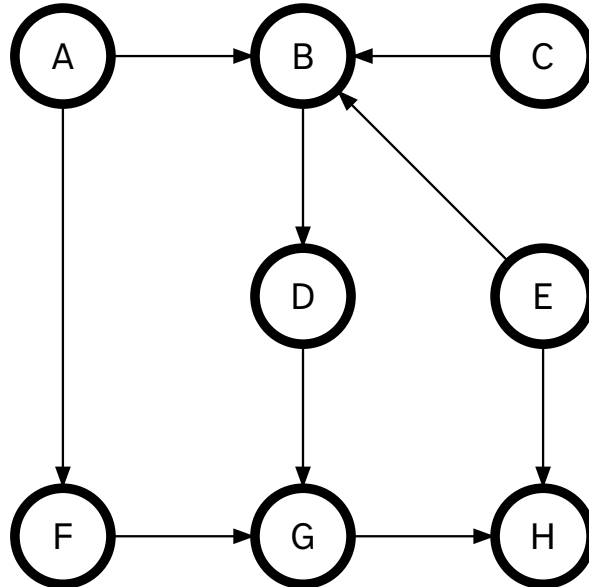


CIS 5210 – Bayes Nets Practice

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Problem 1: Simplifying a Joint Distribution



Assuming that all of the random variables $A - H$ are boolean-valued, a joint distribution $P(A, B, C, D, E, F, G, H)$ would ordinarily require $2^8 - 1$ values to express in total. (There are 2^8 possible assignments, but we know they must sum to 1, so it suffices to know all but one of them.)

By encoding conditional independence assumptions with the Bayes Net above, we can reduce the number of values we need to know in order to represent the joint distribution.

1.1 Encoding the Joint

Using the conditional independences encoded by the Bayes Net above & the chain rule, write out the full product of the joint distribution $P(A, B, C, D, E, F, G, H)$.

$$\begin{aligned} P(A, B, C, D, E, F, G, H) = & P(A) \times P(C) \times P(E) \\ & \times P(D \mid B) \times P(F \mid A) \\ & \times P(G \mid D, F) \times P(H \mid E, G) \\ & \times P(B \mid A, C, E) \end{aligned}$$

1.2 Requirements for a Conditional Probability Table

A single boolean variable conditioned on nothing, e.g. $P(A)$ can be expressed with the following table:

A	$P(A)$
a	0.3
$\neg a$	0.7

Although this table contains two numbers, one of them is redundant. We would say that $P(A)$ can be represented using just a single value, then.

A conditional distribution is a little trickier. Take, for example, $P(D \mid B)$:

B	D	$P(D \mid B)$
b	d	0.9
b	$\neg d$	0.1
$\neg b$	d	0.2
$\neg b$	$\neg d$	0.8

The table has four rows, but it's actually more like the concatenation of two different probability distributions:

D	$P(D \mid b)$
d	0.9
$\neg d$	0.1

D	$P(D \mid \neg b)$
d	0.2
$\neg d$	0.8

Each of these sub-distributions can be represented with a single number, and so $P(D \mid B)$ can really be represented with just two numbers. That brings the total numbers required to store the entire conditional probability table for the Bayes Net above up to three numbers so far.

How many numbers are required in total to represent $P(A, B, C, D, E, F, G, H)$?

$$\begin{aligned}
 P(A, B, C, D, E, F, G, H) &= P(A) \times P(C) \times P(E) \\
 &\quad \times P(D \mid B) \times P(F \mid A) \\
 &\quad \times P(G \mid D, F) \times P(H \mid E, G) \\
 &\quad \times P(B \mid A, C, E) \\
 &\rightarrow 1 + 1 + 1 \\
 &\quad + 2 + 2 \\
 &\quad + 4 + 4 \\
 &\quad + 8 \\
 &= 23
 \end{aligned}$$